

Celebal Excellence Internship - Assignment Week – 4

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

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Objective

To understand Azure cloud concepts and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory.

1. Resource Group Creation

Steps

1. Open Azure Portal.
2. Search for **Resource Groups**.
3. Click **Create**.
4. Enter Resource Group Name: CEI_week4_rg
5. Select Region: Central India
6. Leave Tags empty.
7. Click **Review + Create**.
8. Click **Create**.

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

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CVR COLLEGE OF ENGINEERING L...

Home > Resource Manager

Resource Manager | Resource groups

Summarize my costs by service

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Search

+ Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group by none

Filter for any field...

Subscription equals all

Location equals all

Add filter

Name	Subscription	Location
ADF_Practice	Azure for Students	East US
CEI_week4_rg	Azure for Students	Central India

Showing 1 - 2 of 2. Display count: auto

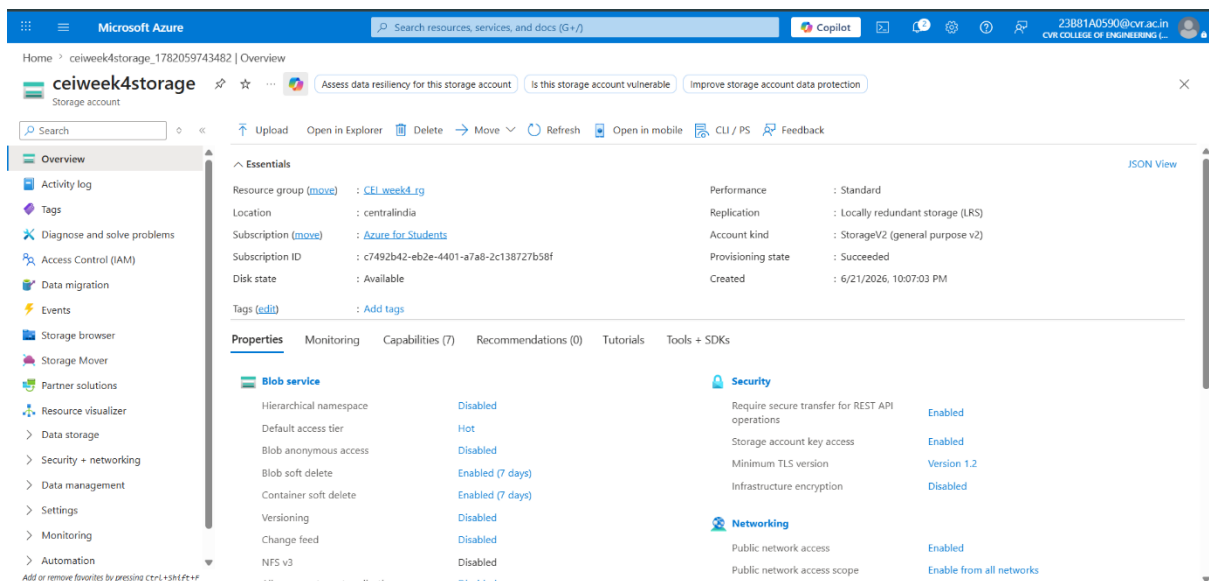
Add or remove favorites by pressing ctrl+shift+F

Give feedback

2. Storage Account Creation

Steps

1. Search for **Storage Accounts**.
2. Click **Create**.
3. Select Resource Group: CEI_week4_rg
4. Enter Storage Account Name: ceiweek4storage
5. Select Region: Central India
6. Performance: Standard
7. Redundancy: LRS
8. Click **Review + Create**.
9. Click **Create**.



3. Blob Container Creation

Steps

1. Open Storage Account.
2. Click **Containers**.
3. Click **+ Container**.
4. Enter Container Name: superstore-data
5. Access Level: Private

6. Click Create.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area is titled 'ceiweek4storage | Containers'. On the left, there is a sidebar with various navigation options, including 'Containers'. The main area displays a table of containers. The table has columns for 'Name', 'Last modified', 'Anonymous access level', 'Lease state', and 'Storage connector'. Two containers are listed: '\$logs' and 'superstore-data'. The 'superstore-data' container is selected, and its details are shown in the right pane.

Name	Last modified	Anonymous access level	Lease state	Storage connector
\$logs	6/21/2026, 10:07:26 PM	Private	Available	-
superstore-data	6/21/2026, 10:23:28 PM	Private	Available	-

4. Upload CSV File

Steps

1. Open superstore-data.
2. Click Upload.
3. Select Sample - Superstore.csv.
4. Click Upload.

The screenshot shows the Microsoft Azure portal interface for the 'superstore-data' container. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area is titled 'superstore-data | Overview'. On the left, there is a sidebar with various navigation options, including 'Overview'. The main area displays the details of the 'superstore-data' container. The 'Overview' tab is selected, and the 'Sample - Superstore.csv' file is listed in the table. The table has columns for 'Name', 'Last modified', 'Access tier', 'Blob type', 'Size', and 'Lease state'. One file is listed: 'Sample - Superstore.csv'.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/21/2026, 10:26:01 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

5. Azure Data Factory Creation

Steps

1. Search for **Data Factory**.
2. Click **Create**.
3. Select Resource Group: CEI_week4_rg
4. Enter Name: nithin-cei-week4-adf
5. Select Region: Central India
6. Select Version: V2
7. Click **Review + Create**.
8. Click **Create**.

The screenshot displays the Microsoft Azure portal interface. At the top, the header shows 'Microsoft Azure' with a search bar and user profile information. The main content area is titled 'CreateDataFactory-20260621223904 | Overview'. A sidebar on the left contains navigation links for 'Overview', 'Inputs', 'Outputs', and 'Template'. The main panel shows a green checkmark icon and the text 'Your deployment is complete'. Below this, deployment details are listed: 'Deployment name : CreateDataFactory-20260621223904', 'Subscription : Azure for Students', and 'Resource group : CEI_week4_rg'. A table titled 'Deployment details' shows the resource 'nithin-cei-week4-adf' of type 'Data factory (V2)' with a status of 'OK'. On the right side, there are three promotional cards: 'Cost management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. At the bottom left, a small note reads 'Add or remove favorites by pressing Ctrl+Shift+F'.

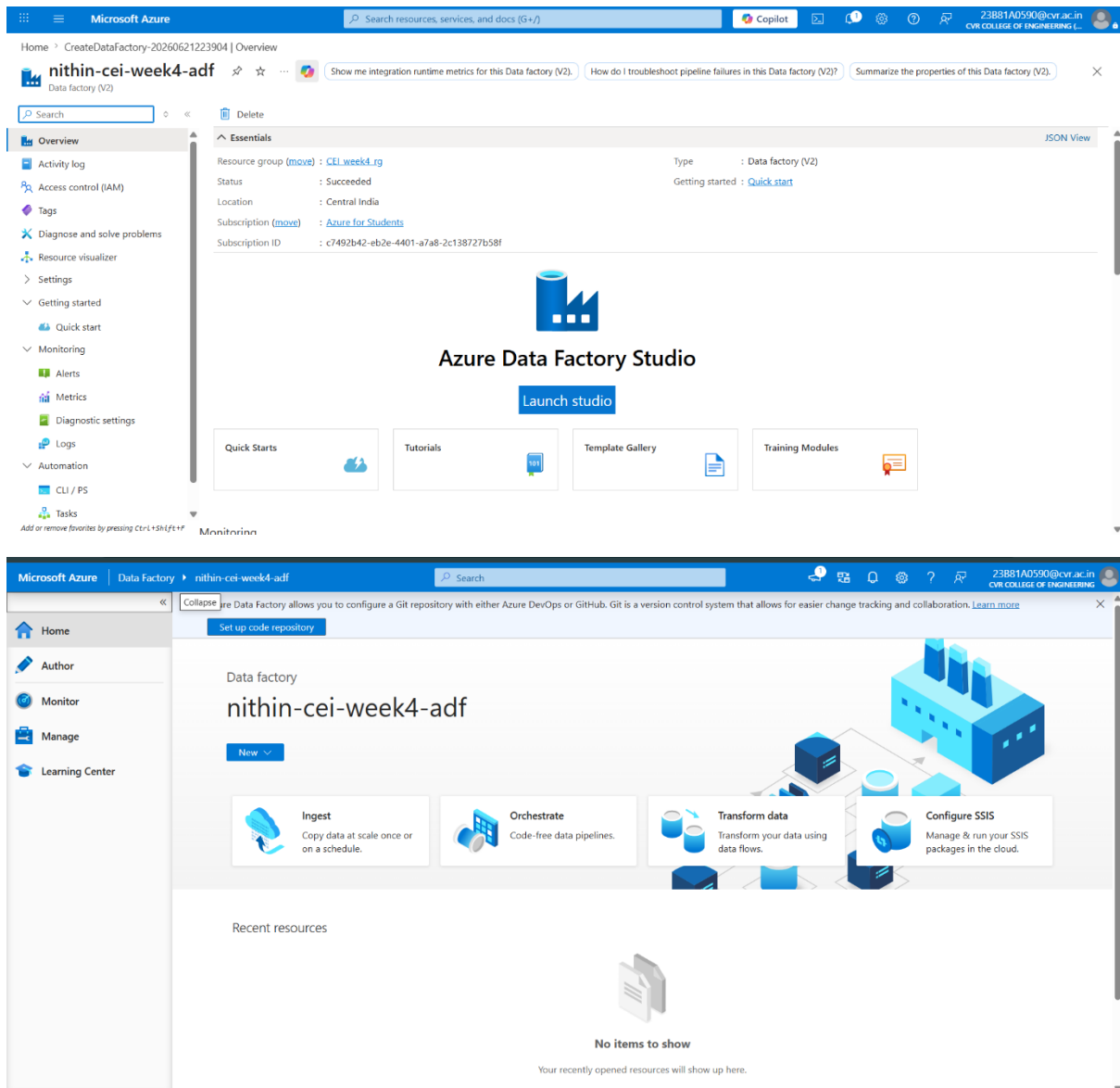
Resource	Type	Status	Operation details
nithin-cei-week4-adf	Data factory (V2)	OK	Operation details

6. Explore ADF Studio

Steps

1. Open Azure Data Factory.
2. Click **Launch Studio**.
3. Observe:
 - Author

- Monitor
- Manage

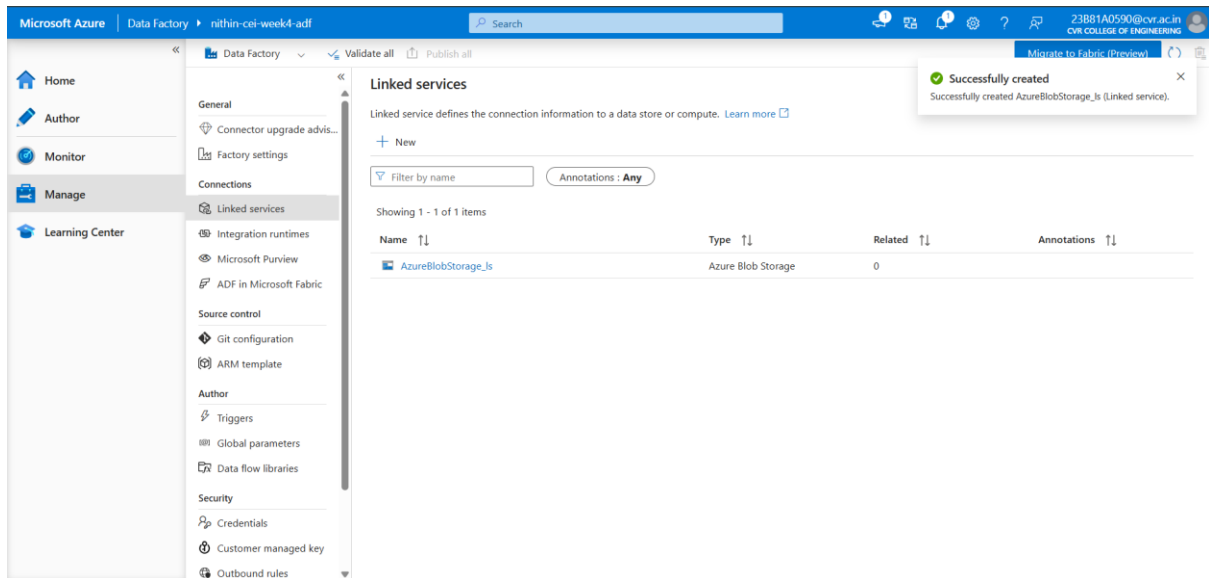


7. Create Linked Service

Steps

1. Open **Manage**.
2. Click **Linked Services**.
3. Click **+ New**.
4. Select **Azure Blob Storage**.
5. Click **Continue**.

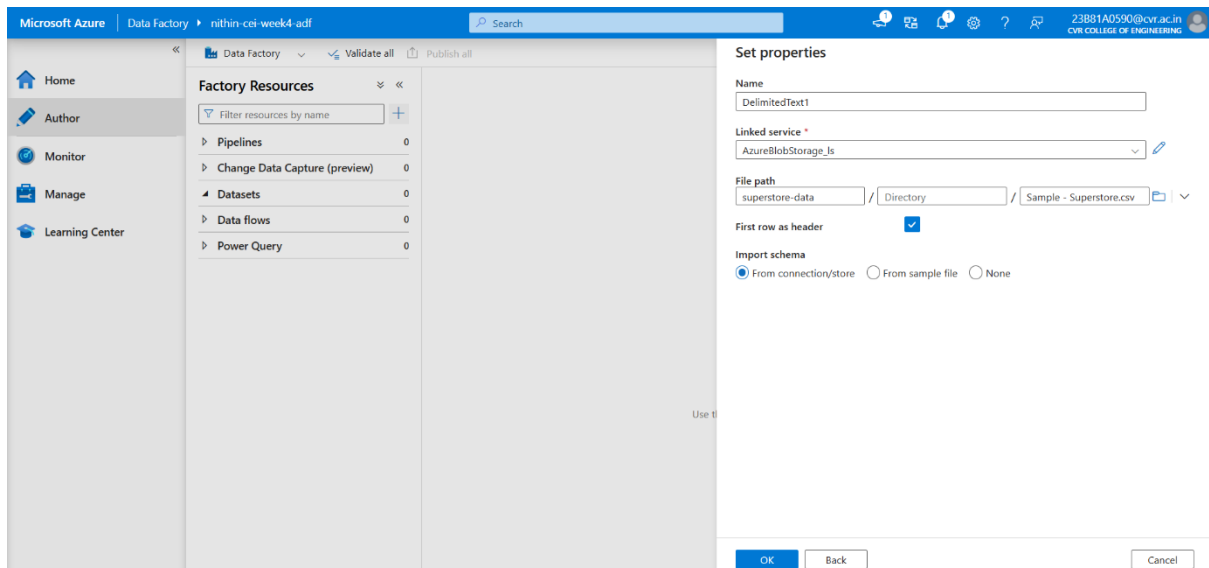
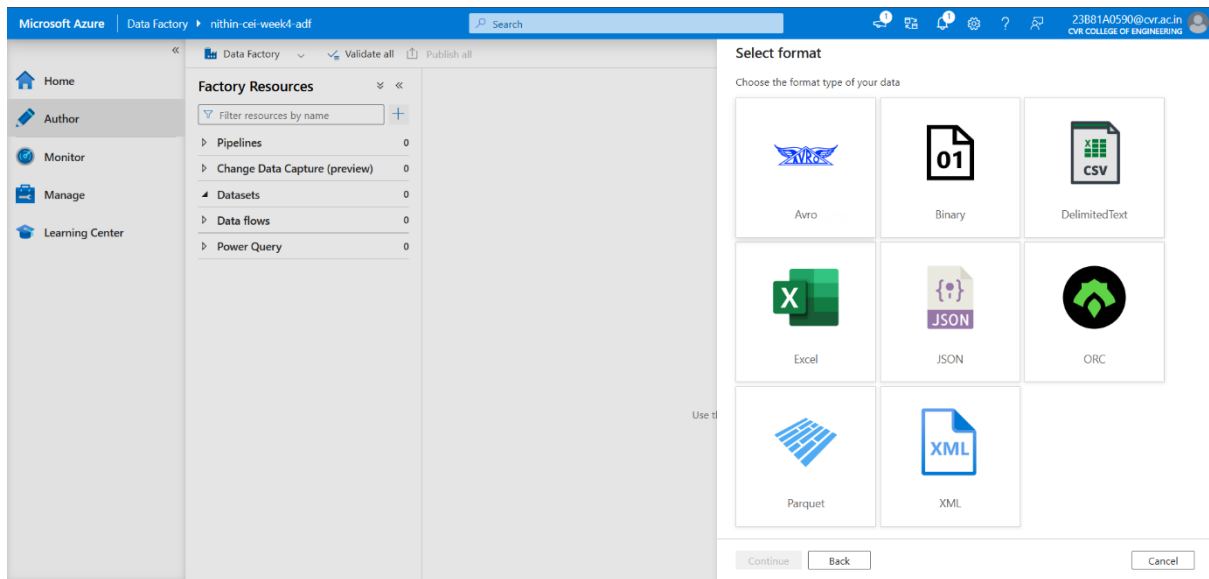
6. Enter Name: AzureBlobStorage_ls
7. Select Storage Account: ceiweek4storage
8. Click **Test Connection**.
9. Click **Create**.



8. Create Source Dataset

Steps

1. Open **Author**.
2. Click **+**.
3. Click **Dataset**.
4. Select **Azure Blob Storage**.
5. Select **Delimited Text**.
6. Enter Name: Superstore_Source_DS
7. Select Linked Service: AzureBlobStorage_ls
8. Select Container: superstore-data
9. Select File: Sample - Superstore.csv
10. Click **OK**.



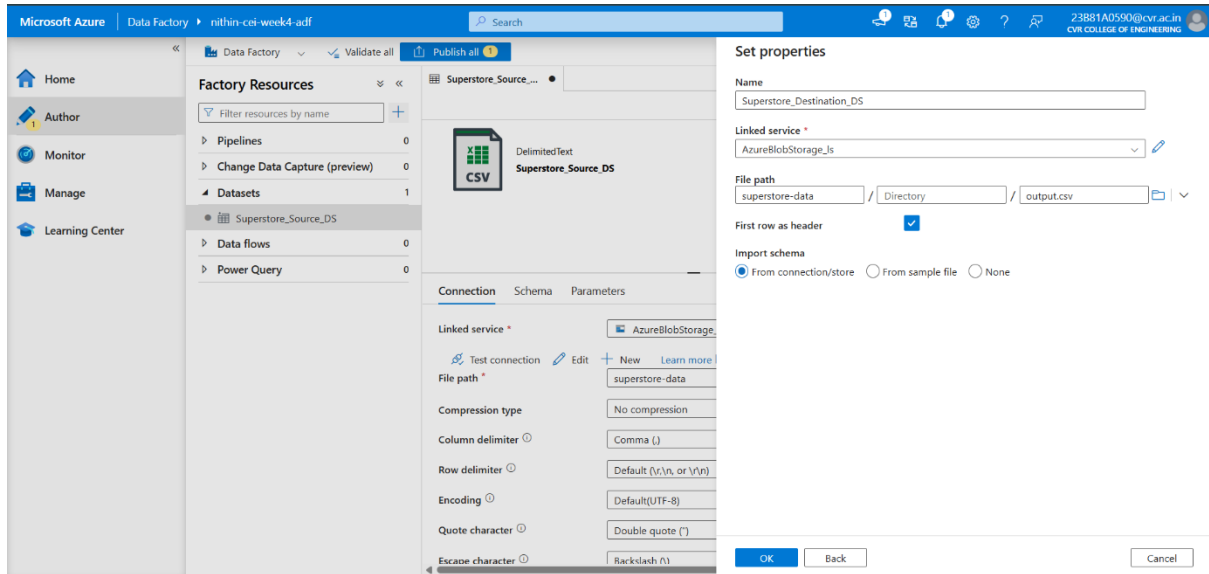
9. Create Destination Dataset

Steps

1. Click +.
2. Click **Dataset**.
3. Select **Azure Blob Storage**.
4. Select **Delimited Text**.
5. Enter Name: Superstore_Destination_DS
6. Select Linked Service: AzureBlobStorage_ls
7. Select Container: superstore-data
8. Enter File Name: output.csv

9. Import Schema: None

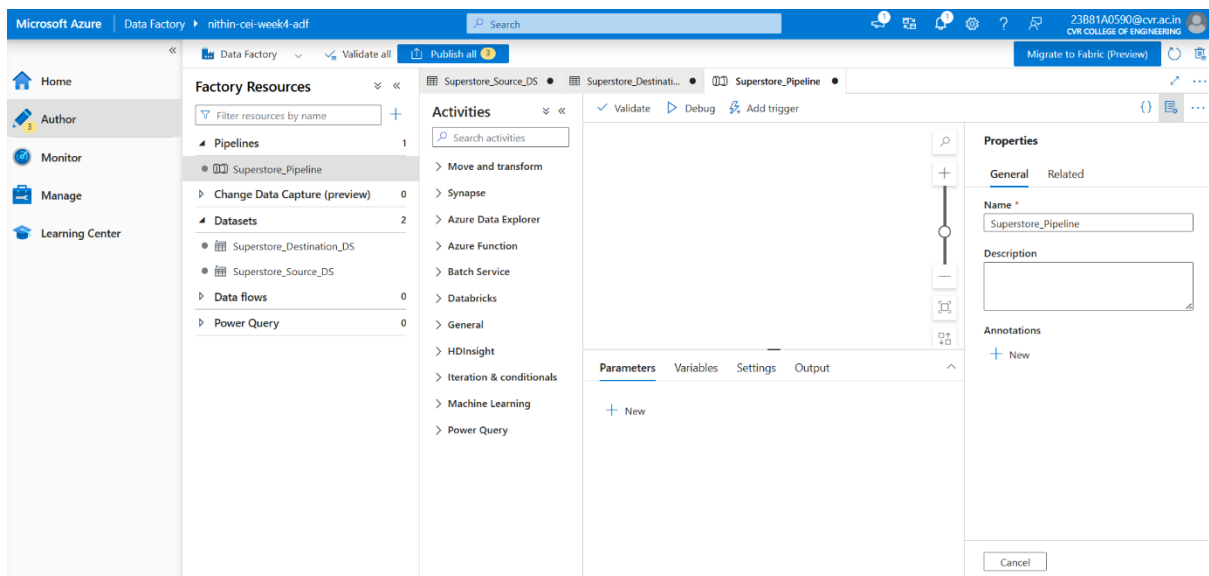
10. Click OK.



10. Create Pipeline

Steps

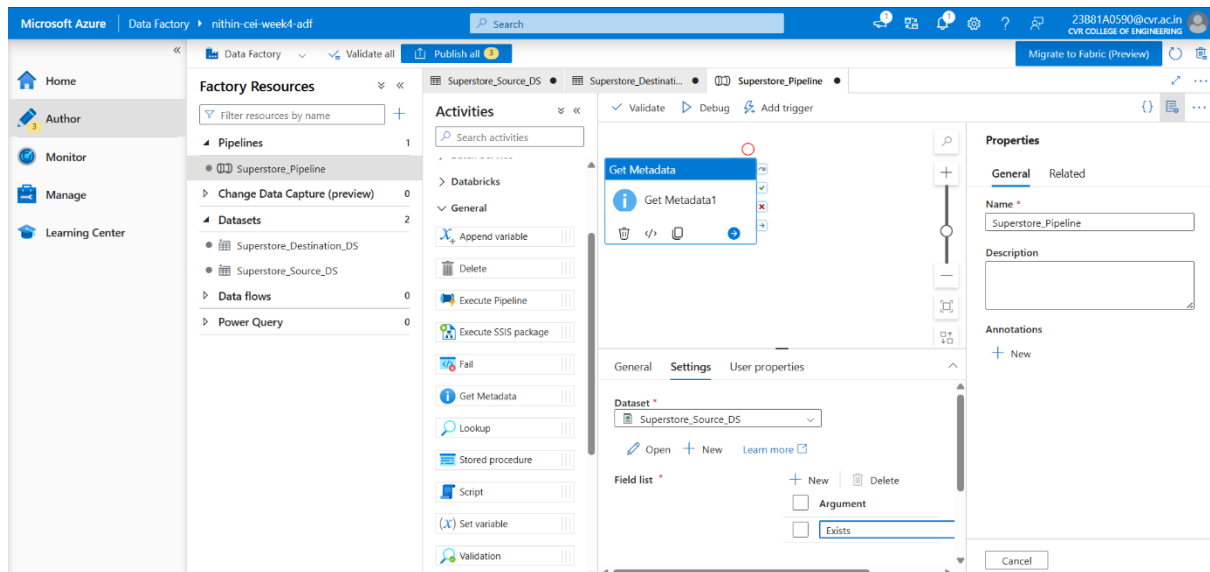
1. Click +.
2. Click **Pipeline**.
3. Enter Pipeline Name: Superstore_Pipeline



11. Configure Get Metadata Activity

Steps

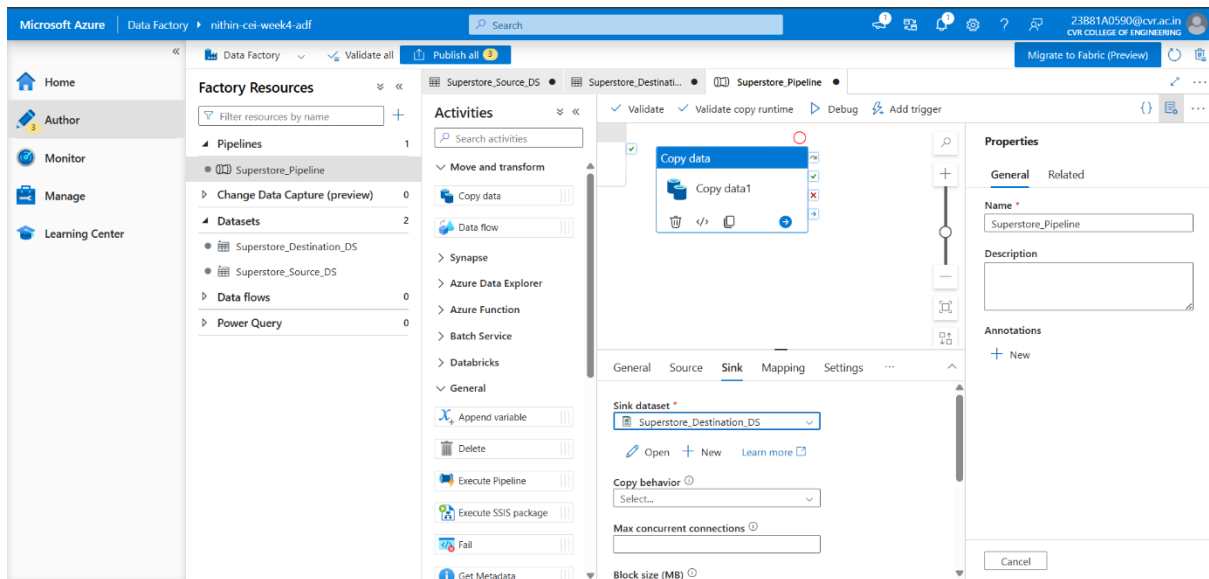
1. Drag **Get Metadata** activity.
2. Select Dataset: Superstore_Source_DS
3. Open Settings.
4. Field List → Exists



12. Configure Copy Data Activity

Steps

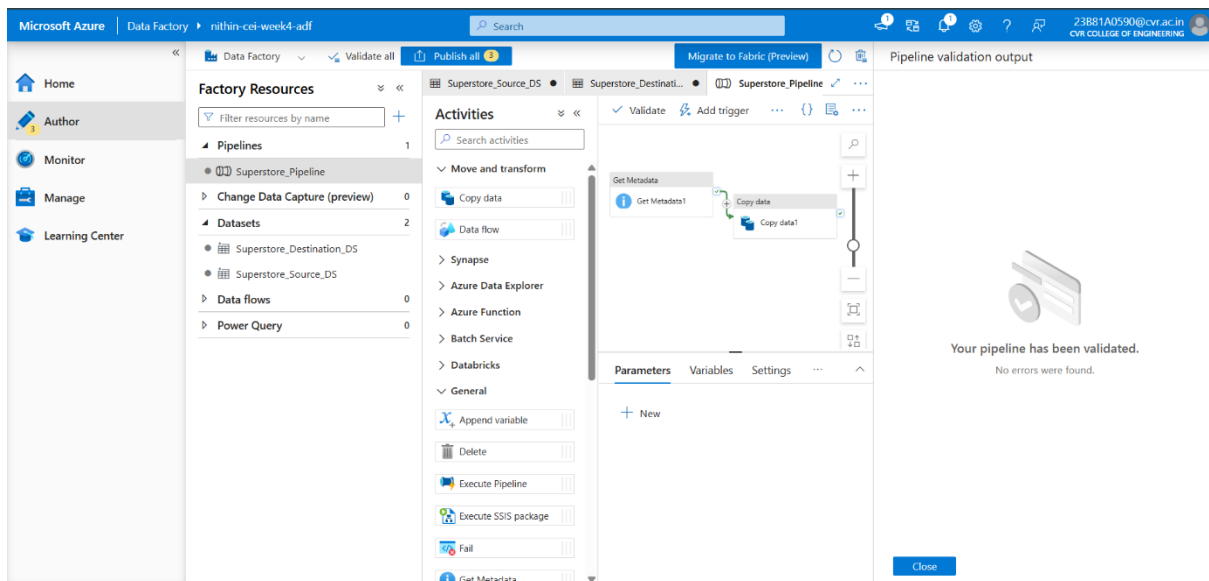
1. Drag **Copy Data** activity.
2. Connect **Get Metadata** → **Copy Data**
3. Open Source Tab.
4. Select Dataset: Superstore_Source_DS
5. Open Sink Tab.
6. Select Dataset: Superstore_Destination_DS



13. Validate Pipeline

Steps

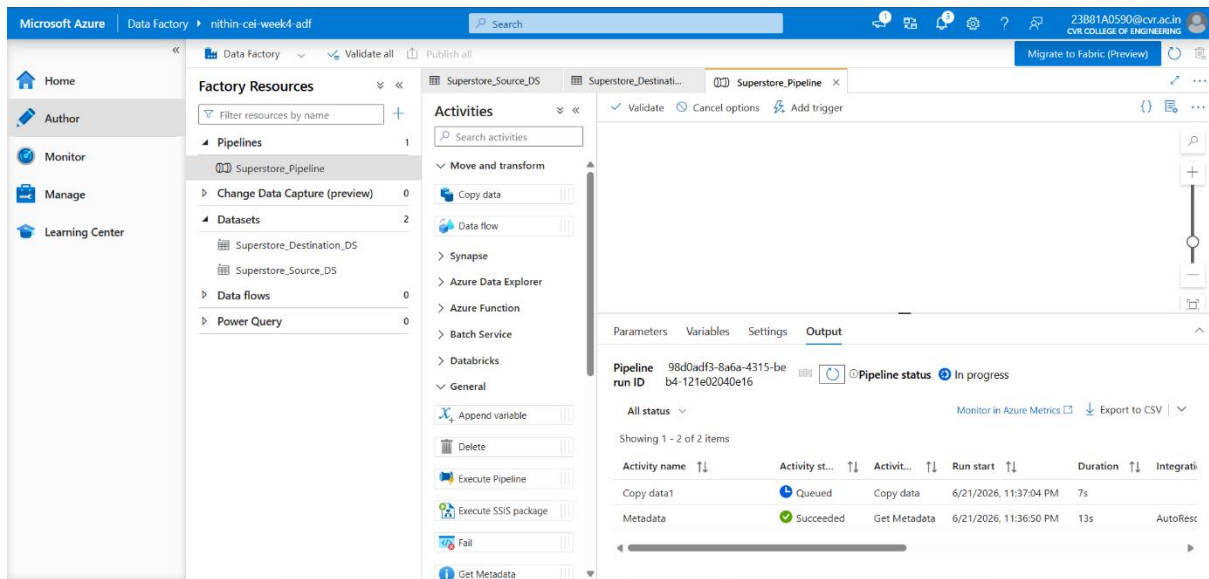
1. Click **Validate All**.
2. Verify validation completed successfully.



14. Publish Pipeline

Steps

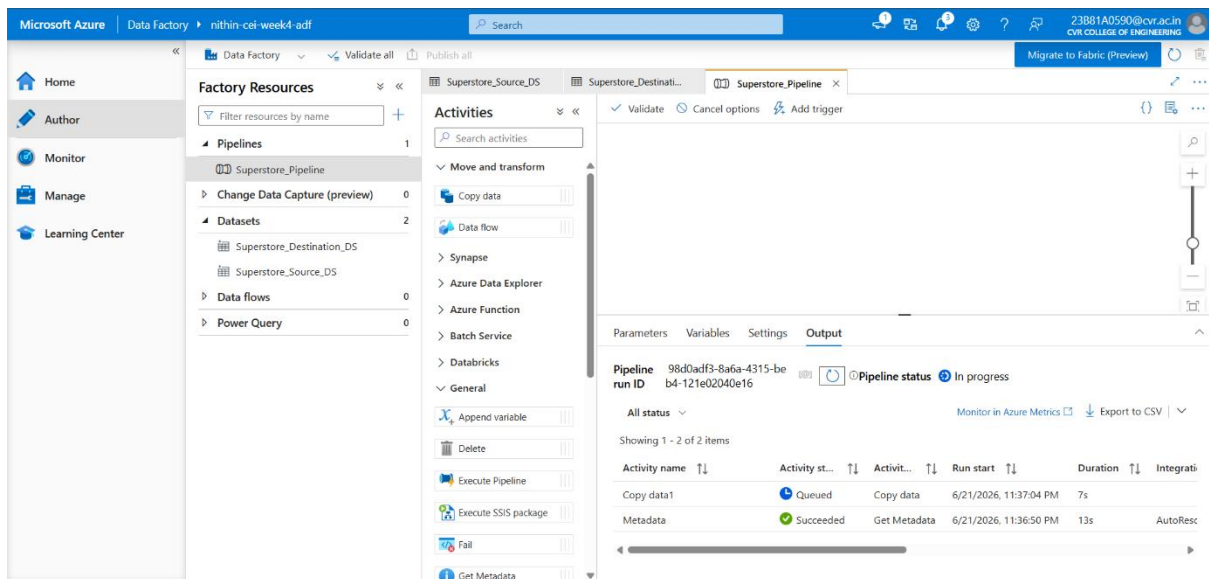
1. Click **Publish All**.
2. Click **Publish**.



15. Execute Pipeline

Steps

1. Click **Debug**.
2. Wait for pipeline execution.



16. Monitor Pipeline Execution

Steps

1. Open **Monitor**.
2. Verify:

- Metadata → Succeeded
- Copy Data → Succeeded
- Pipeline → Succeeded

The screenshot displays the Microsoft Azure Data Factory Author interface. The left sidebar shows the navigation menu with 'Author' selected. The main workspace is divided into three panes: 'Factory Resources', 'Activities', and 'Output'.

Factory Resources: Shows a list of resources including Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Superstore_Pipeline' is selected.

Activities: A list of activities is shown, including 'Copy data', 'Data flow', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'Append variable', 'Delete', 'Execute Pipeline', 'Execute SSIS package', 'Fail', and 'Get Metadata'.

Output: The 'Output' tab is active, showing the pipeline status as 'Succeeded'. The pipeline ID is '98d0adf3-6afa-4315-beb4-121e02040e16'. The output table shows the following data:

Activity name	Activity st...	Activit...	Run start	Duration	Integrati
Copy data1	Succeeded	Copy data	6/21/2026, 11:37:04 PM	15s	AutoResc
Metadata	Succeeded	Get Metadata	6/21/2026, 11:36:50 PM	13s	AutoResc

17. Verify Output File

Steps

1. Open Storage Account.
2. Open Container: superstore-data
3. Verify output.csv is created.

The screenshot shows the Microsoft Azure Storage Explorer interface. The left sidebar displays the 'superstore-data' container. The main workspace shows the 'Overview' tab for the container, which lists two blobs:

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/21/2026, 10:26:01 PM	Hot (Inferred)	Block blob	2.18 MiB	Available
output.csv	6/21/2026, 11:37:16 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

18. IAM Roles

Steps

1. Open Storage Account.
2. Open **Access Control (IAM)**.
3. Open **Role Assignments** tab.
4. Open **Roles** tab.
5. Verify:
 - Owner
 - Contributor
 - Reader

Microsoft Azure

Home > ceiweek4storage

ceiweek4storage | Access Control (IAM)

Storage account

Search

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables
Security + networking

Check access | **Role assignments** | Roles | Deny assignments

Number of role assignments for this subscription: 2 / 4000

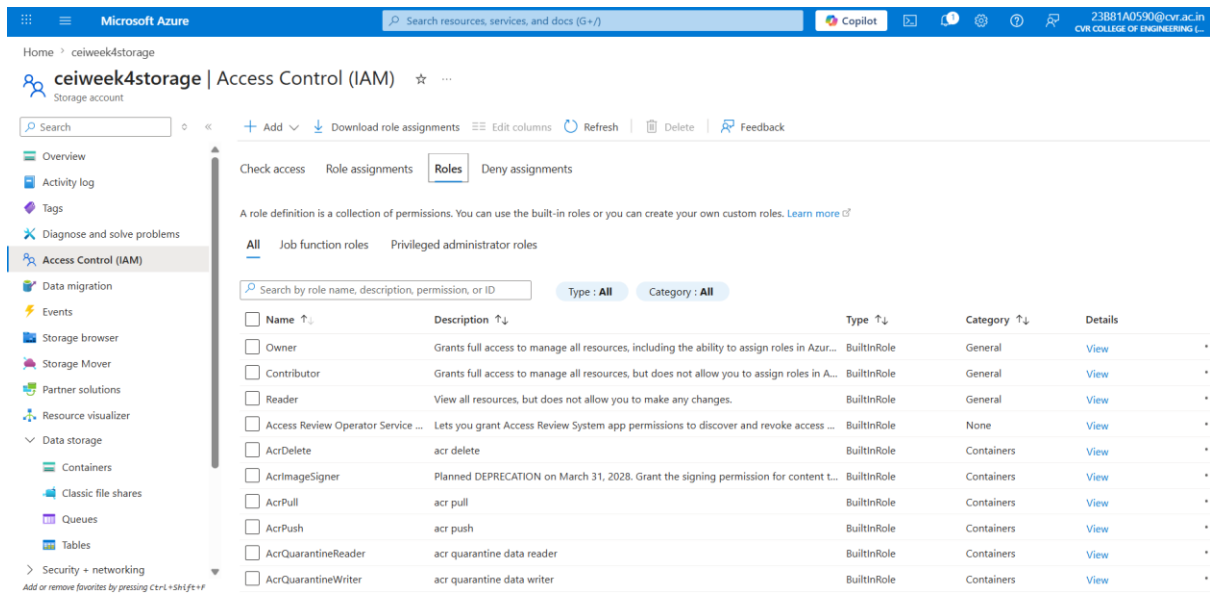
Search by name or email

Type: All | Role: All | Scope: All scopes | Group by: Role

All (2) | Job function roles (0) | Privileged administrator roles (2)

☐	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
☒	Owner (2)				
☒	SIGA NITHIN 23B81A0590@cvr.ac.in	User	Owner	Subscription (Inherited)	None
☒	SIGA NITHIN 23B81A0590@cvr.ac.in	User	Owner	Subscription (Inherited)	None

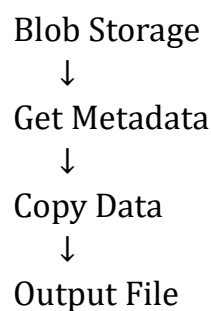
Showing 1 - 2 of 2 results.



Explored IAM and reviewed Owner, Contributor, and Reader roles. Role assignment changes were not available in the current subscription.

Learnings :

- Learned how Azure resources are organized using Resource Groups.
- Understood the purpose of Storage Accounts and Blob Containers for storing files in Azure.
- Explored Azure Data Factory (ADF) and its main modules: Author, Monitor, and Manage.
- Created a Linked Service to establish connectivity between ADF and Blob Storage.
- Learned how Datasets represent source and destination data in a pipeline.
- Used Get Metadata activity to validate the existence of the source file before processing.
- Used Copy Data activity to move data from source to destination.
- Understood the complete pipeline flow:



- Learned how to validate, publish, execute, and monitor pipelines in Azure Data Factory.
- Verified successful data movement by generating output.csv.
- Explored IAM (Identity and Access Management) and understood the purpose of Owner, Contributor, and Reader roles.
- Gained hands-on experience in building an end-to-end cloud-based data pipeline using Azure services.

Conclusion :

An end-to-end data pipeline was successfully created using Azure Blob Storage and Azure Data Factory. The source CSV file was validated using Get Metadata activity and copied successfully to the destination file using Copy Data activity. Pipeline execution, monitoring, output verification, and IAM role exploration were completed successfully.